

RTTI — Temporary traffic control measures

Introduction to semantic modelling

A **concept scheme** (in SKOS: *ConceptScheme*) is a structured collection of concepts that carry a fixed, shared meaning within a particular domain. The goal is for everyone working with the model — people and systems alike — to use the same words for the same things. That sounds obvious, but in practice language causes a surprising amount of confusion: what one organisation calls a *measurement point*, another calls a *sensor*, a *location*, or an *observation station*.

A concept scheme solves this by recording, for each concept:

- a **preferred label** (`prefLabel`) it is known by,
- a **definition** that unambiguously describes what the concept does and does not cover,
- and optionally a **scope note** or **example** to clarify borderline cases.

Concepts rarely stand on their own. They are related **hierarchically** (a vehicle is broader than a truck) or **associatively** (a road user relates to a road segment). These relationships make it possible for queries and analyses to navigate between general and specific concepts, and to make connections between concepts explicit. Within SKOS we use `broader`, `narrower`, and `related` for that.

When a concept also exists elsewhere — in a national or European standard, for instance — we record that as a **mapping relation**. This prevents reinventing the wheel: we link to the source so interoperability with external systems remains possible.

This document describes the concept scheme as it is currently recorded. The following sections list the metadata, every concept, the relationships between them, and — where the model is connected to a data source — how it is linked to that source data.

RTTI — Temporary traffic control measures

Description

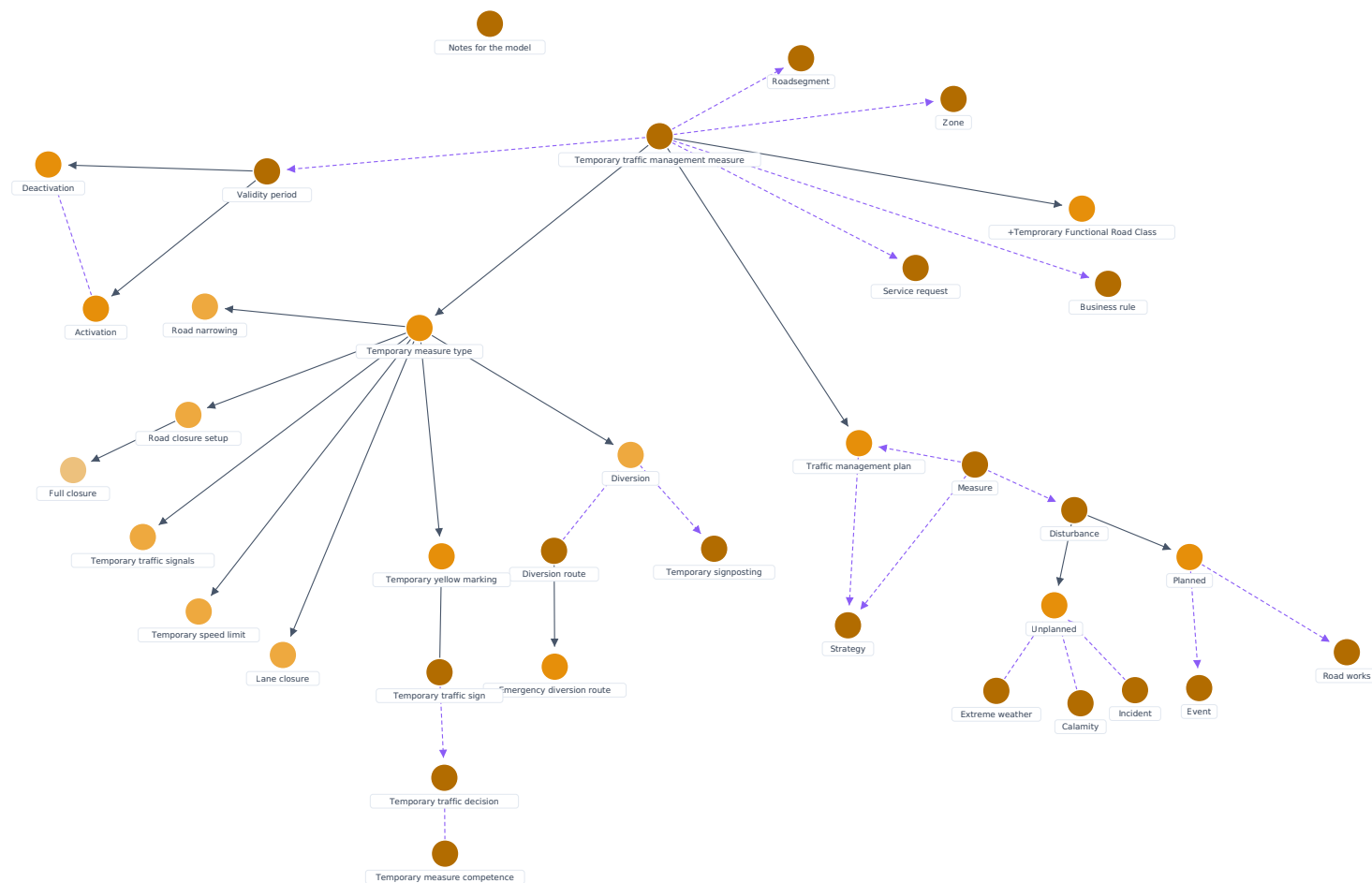
This concept scheme describes the set of temporary traffic management measures taken by or on behalf of the road authority (for a bounded validity period), which is activated for and deactivated after one or more specific triggers or scenarios.

Metadata

Field	Value
URI	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen
Version	0.9.1
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Organization	Nationaal Toegangspunt Mobiliteit
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Last modified by	arthur.keesen@ndw.nu

Model overview

Visual overview of the concepts and their relations. Hierarchical relations are drawn with solid arrows; associative relations are drawn with dashed arrows.



—> broader / narrower (hiërarchisch / Associatief)
 -.-> related (gerelateerd)
 Bron: Ontologica — RTTI — Tijdelijke verkeersbeheersmaatregelen

Concepts

Concepts overview

Concept	Definition	Scope	Source
Temporary traffic management measure	A set of temporary traffic management measures adopted by or on behalf of the road authority (for a limited period of validity), which is activated before and deactivated after one or more specific disruptions.	Distinguished from a permanent traffic circulation plan by the activation and deactivation characteristic: a temporary measure is deactivated as soon as the triggered scenario has been resolved (NAPCORE data dictionary).	https://eur-lex.europa.eu/eli/reg_del/2022/670/oj , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf , https://wetten.overheid.nl/BWBR0004826
Temporary traffic decision	A traffic decision underlying a temporary placement or temporary measure, required where the circumstances last a few months or occur on a regular basis.	WVW 1994 in conjunction with BABW art. 37: by way of derogation from the main rule, temporary placement and temporary measures are effected pursuant to a traffic decision where the circumstances last longer than four months or occur on a regular basis. Case law (including the Council of State) allows a traffic decision to be waived in the case of urgent circumstances of a passing nature.	https://wetten.overheid.nl/BWBR0004826 , https://wetten.overheid.nl/BWBR0006622
Temporary measure competence	The competence of the road authority to place temporary traffic signs or take temporary measures in the case of an urgent circumstance of a passing nature, without an underlying traffic decision.	BABW art. 34 (urgent circumstance of a passing nature, for the duration of that circumstance) and art. 35 (placement/measure may be effected without a traffic decision).	https://wetten.overheid.nl/BWBR0004826
Temporary traffic sign	A traffic sign that is placed or applied on a temporary basis and is removed or deactivated once the trigger has ended.	RWV 1990: temporarily placed or applied road surface traffic signs take precedence over other road surface markings applied at that location in so far as these are incompatible (art. 63a).	https://wetten.overheid.nl/BWBR0004825 , https://wetten.overheid.nl/BWBR0004826
Road works	Maintenance, construction or repair works on or alongside a road section that wholly or partly affect the use of the road.	NAPCORE data dictionary: road works generally affect individual lanes; where they affect all lanes, they coincide with road closures. In the Netherlands supported by the NDW DATEX II v3 road works profile.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf , https://wetten.overheid.nl/BWBR0004826

Concept	Definition	Scope	Source
			www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/
Temporary measure type	A category of physical or regulatory intervention by which a traffic measure is implemented with a clear start time and end time.	Collective concept for closures, lane measures, speed measures and temporary traffic signals, in line with the sub-measures in the NDW DATEX II v3 profile.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/
Road closure setup	The physical closing off of a work zone or road section using delineation, barrier equipment and markings.	Elaborated in the CROW work in progress guidelines (96a/96b); terms reusable from the CROW thesaurus.	https://data.crow.nl/thesaurus/ , https://www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/
Full closure	The closure of a road section to all traffic, possibly with the exception of emergency services.	Modelled in the NDW DATEX II v3 profile via RoadOrCarriagewayOrLaneManagement; where the closure affects all lanes it coincides with a road closure (NAPCORE).	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Lane closure	The temporary closure of one or more lanes while the remaining lanes stay available.	RTTI data category lane closures (planned/unplanned); in the NDW profile via RoadOrCarriagewayOrLaneManagement.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Road narrowing	The temporary narrowing or shifting of the carriageway or lane for the purpose of works or another scenario.	Part of lane and carriageway management for road works in the NDW DATEX II v3 profile.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/
Temporary speed limit	A temporarily reduced speed limit imposed during road works or another scenario and lifted once these have ended.	Recorded in DATEX II as SpeedManagement with speedManagementType speedRestrictionInOperation, a temporarySpeedLimit and complianceOption mandatory (NDW profile). References the speed restrictions model without duplicating it.	https://docs.datex2.eu/ , https://docs.ndw.nu/dataformaten/datex2-v3/wwz/

Concept	Definition	Scope	Source
Temporary traffic signals	A temporarily installed traffic signal control installation, for example to regulate traffic past a work zone.	In the NDW DATEX II v3 profile via GeneralNetworkManagement with generalNetworkManagementType temporaryTrafficLights.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/
Temporary yellow marking	A marking applied temporarily to the road surface that takes precedence over the permanent white marking.	RVV 1990 art. 63a governs the precedence of temporary road surface signs; the colour requirement (a colour other than white, in practice yellow) follows from the BABW implementing regulations, Chapter IV, section 1. References the traffic signs model without duplication.	https://wetten.overheid.nl/BWBR0004825 , https://wetten.overheid.nl/BWBR0004826
Diversion	The temporary guidance of traffic along an alternative route because of a closure or restriction on the usual route.	Modelled in DATEX II via ReroutingManagement; in the Netherlands elaborated in the CROW work in progress guidelines.	https://docs.datex2.eu/ , https://www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/
Diversion route	The alternative route that traffic follows for as long as the usual route is closed or restricted.	Part of the diversion design according to the CROW work in progress guidelines.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/
Temporary signposting	Temporary indications on or next to the existing signposting that indicate the diversion route.	Designed using yellow diversion signs in accordance with the CROW work in progress guidelines.	https://www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/
Emergency diversion route	A predefined and numbered diversion route (Dutch: U-route) that is opened when a motorway is closed.	The Dutch U-route system is managed by Rijkswaterstaat and indicated by U-signs.	https://www.crow.nl/Onderwerpen/Arbo-en-Veiligheid/Werk-in-uitvoering/ , https://www.rijkswaterstaat.nl/
Disturbance	The event or condition that causes the activation of a set off temporary traffic measures or a strategy.	Collective concept for the situations that trigger a traffic management plan or a temporary measure (NAPCORE data dictionary).	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Event		Placed under road works and events in the NDW DATEX II v3 profile.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://napcore.eu/

Concept	Definition	Scope	Source
	A planned occurrence, such as a sports fixture, market or festival, that requires temporary traffic measures.		wp-content/uploads/2025/05/DD_2022_670.pdf
Calamity	An unexpected high-impact event, such as a natural or technological disaster, that requires temporary measures.	Falls under the scenario-driven measures activated through a traffic management plan (NAPCORE data dictionary).	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Incident	An unexpected disruption of the traffic flow, such as an accident or a broken-down vehicle.	RTTI state-of-the-network category; triggers temporary measures and the provision of information.	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Extreme weather	Weather conditions, such as storm, black ice or snow, that require the activation of a set of temporary traffic measures.	One of the conditions that can trigger a traffic management plan (NAPCORE data dictionary).	https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Traffic management plan	The collection of strategies, measures and actions that together constitute the plan.	The traffic management plan is a defined DATEX II data type (TrafficManagementPlan namespace, CEN/TS 16157-8) with the classes Strategy and Measure carrying activation conditions; unlike the traffic circulation plan, which has no DATEX II namespace of its own.	https://docs.datex2.eu/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Strategy	A coherent set of measures within a traffic management plan sharing a common activation condition.	Modelled in DATEX II as Strategy within the TrafficManagementPlan namespace; activation conditions may be composite (v3.6).	https://docs.datex2.eu/
Measure	An individual traffic control measure within a traffic management plan, with its own activation condition.	Modelled in DATEX II as Measure within the TrafficManagementPlan namespace.	https://docs.datex2.eu/
Validity period	The start and end time within which a temporary traffic measure is in force.	The characteristic that distinguishes temporary measures from permanent ones; recorded in DATEX II and the NDW profile as the validity period.	https://docs.datex2.eu/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
Activation		In the NDW chain linked to the active reporting of a measure.	

Concept	Definition	Scope	Source
	The entry into force of a temporary traffic measure upon occurrence of the trigger.		https://docs.datex2.eu/ , https://docs.ndw.nu/dataformaten/datex2-v3/wwz/
Deactivation	The termination of a temporary traffic measure once the scenario that triggered the measure has been resolved.	Core distinction between the traffic management plan and the traffic circulation plan according to the NAPCORE data dictionary; in the NDW chain the transition from active to idle status, whereby the associated measures are deactivated.	https://docs.ndw.nu/dataformaten/datex2-v3/wwz/ , https://napcore.eu/wp-content/uploads/2025/05/DD_2022_670.pdf
+Temporary Functional Road Class	A temporarily assigned functional road class that overrides the normal classification of a road section for as long as a temporary traffic management measure is in force.		
Zone	A contiguous geographic area within which a temporary traffic management measure or traffic management plan applies. Unlike a road section, which denotes a linear stretch of road, a zone delimits an areal region to which the relevant measures, rules or conditions apply.		
Business rule	Een expliciet vastgelegde regel die bepaalt onder welke condities een tijdelijke verkeersbeheersmaatregel wordt geactiveerd, gewijzigd of gedeactiveerd.		
Notes for the model	Voor validatie doeleinden		
Roadsegment	T.b.d.		
Service request	t.b.d.		

Concept	Definition	Scope	Source
Planned	A disruption that is known and scheduled in advance.		
Unplanned	A disruption that is not scheduled in advance and occurs unexpectedly.		

Relations per concept

Outgoing relations only. Semantic relations stay inside the model; mapping relations link to concepts elsewhere.

Temporary traffic management measure

Semantic relations

Type	Target label	Target concept
narrower	+Temporary Functional Road Class	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1787920015950
narrower	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301
narrower	Traffic management plan	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000601
related	Zone	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1787920171026
related	Business rule	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1787920279579
related	Roadsegment	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1787920438369
related	Service request	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1787920547796
related	Validity period	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000701

Temporary measure competence

Semantic relations

Type	Target label	Target concept
related	Temporary traffic decision	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000101

Temporary traffic sign

Semantic relations

Type	Target label	Target concept
narrower	Temporary yellow marking	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000308
related	Temporary traffic decision	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000101

Temporary measure type

Semantic relations

Type	Target label	Target concept
broader	Temporary traffic management measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000001
narrower	Temporary yellow marking	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000308
narrower	Road closure setup	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000302
narrower	Lane closure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000304
narrower	Road narrowing	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000305
narrower	Temporary speed limit	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000306
narrower	Temporary traffic signals	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000307
narrower	Diversion	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000401

Road closure setup

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301
narrower	Full closure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000303

Full closure

Semantic relations

Type	Target label	Target concept
broader	Road closure setup	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000302

Lane closure

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301

Road narrowing

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301

Temporary speed limit

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301

Temporary traffic signals

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301

Temporary yellow marking

Semantic relations

Type	Target label	Target concept
broader	Temporary traffic sign	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000103
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301

Diversion

Semantic relations

Type	Target label	Target concept
broader	Temporary measure type	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000301
related	Diversion route	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000402
related	Temporary signposting	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000403

Diversion route

Semantic relations

Type	Target label	Target concept
narrower	Emergency diversion route	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000404
related	Diversion	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000401

Emergency diversion route

Semantic relations

Type	Target label	Target concept
broader	Diversion route	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000402

Disturbance

Semantic relations

Type	Target label	Target concept
narrower	Planned	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1788511855955
narrower	Unplanned	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1788511877986
related	Measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000603

Calamity

Semantic relations

Type	Target label	Target concept
related	Unplanned	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1788511877986

Incident

Semantic relations

Type	Target label	Target concept
related	Unplanned	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1788511877986

Extreme weather

Semantic relations

Type	Target label	Target concept
related	Unplanned	https://nationaal-toegangspunt-mobiliteit.ontologica.nl/schemes/rtti-tijdelijke-verkeersmaatregelen/concept-1788511877986

Traffic management plan

Semantic relations

Type	Target label	Target concept
broader	Temporary traffic management measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000001
related	Measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000603
related	Strategy	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000602

Strategy

Semantic relations

Type	Target label	Target concept
related	Measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000603

Measure

Semantic relations

Type	Target label	Target concept
related	Disturbance	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000501
related	Traffic management plan	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000601
related	Strategy	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000602

Validity period

Semantic relations

Type	Target label	Target concept
narrower	Deactivation	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000703
narrower	Activation	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000702

Activation

Semantic relations

Type	Target label	Target concept
broader	Validity period	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000701
related	Deactivation	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000703

Deactivation

Semantic relations

Type	Target label	Target concept
broader	Validity period	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000701

+Temporary Functional Road Class

Semantic relations

Type	Target label	Target concept
broader	Temporary traffic management measure	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000001

Planned

Semantic relations

Type	Target label	Target concept
broader	Disturbance	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000501
related	Event	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000502
related	Road works	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000201

Unplanned

Semantic relations

Type	Target label	Target concept
broader	Disturbance	https://gegevenscatalogus.gov/rtti-tijdelijke-verkeersmaatregelen/concept-1773250000501

Introduction to RML mappings

An **information description** (technically: an RML mapping) describes how concrete data — records in a file, rows in a table, fields in an XML message — corresponds to the concepts in the concept scheme above. Without such a description data stands on its own; with it, every piece of data is connected to an unambiguous concept.

The principle is simple: for each data source we define one or more **TriplesMaps**. Each TriplesMap is a recipe that says *"each record in this data source represents an instance of that concept, and these fields are its properties"*. For each field we then specify which concept describes the property (for example `hasStatus`, `liesOnRoadSegment`) and how the field should be read from the source.

The essence for the reader of this document:

- the **TriplesMap table** below shows which data sources are linked to this model,
- the **subject template** shows how a unique URI is constructed for each record,
- the **number of predicates** gives a rough indication of how rich the mapping per source is.

The actual mapping rules (field → concept) are managed inside the Ontologica application itself and can be inspected and edited through the mapping wizard.

Information description

TriplesMaps

Name	Datasource	Subject template	rdf:type	# predicates
carriageway	tijdelijke verkeersmaatregelen afsluitingen xml	https://example.org/carriageway/{carriageway}	—	2
carriageway	tijdelijke verkeersmaatregelen maximum snelheden xml	https://example.org/carriageway/{carriageway}	—	1